

Advanced Machine Learning and Neural Networks CS 767

Class 1

Introduction

Prerequisites

- Goes well beyond what a basic data scientist is expected to know.
 - Training for actual MLE positions
 - Or research in machine learning.
- Will be expected to independently learn to code using the Keras API and TensorFlow.
 - Must be willing to spend at least 6-8 hours on weekly homeworks.
- Will also be expected to know essential differentiation and probability
 - I will review both these topics, but only briefly.

Prerequisites (contd.)

- Must have a good grasp of basic ML concepts
 - Supervised, unsupervised, and reinforcement learning
- At least one regression and one classification algorithm
 - What is regression? What is classification?
 - Linear regression,
 - Polynomial regression, and
 - Logistic regression.
- Essentially, if you haven't taken CS 555, 677, 688, or 699.
 - This course isn't for you.

General Information Document

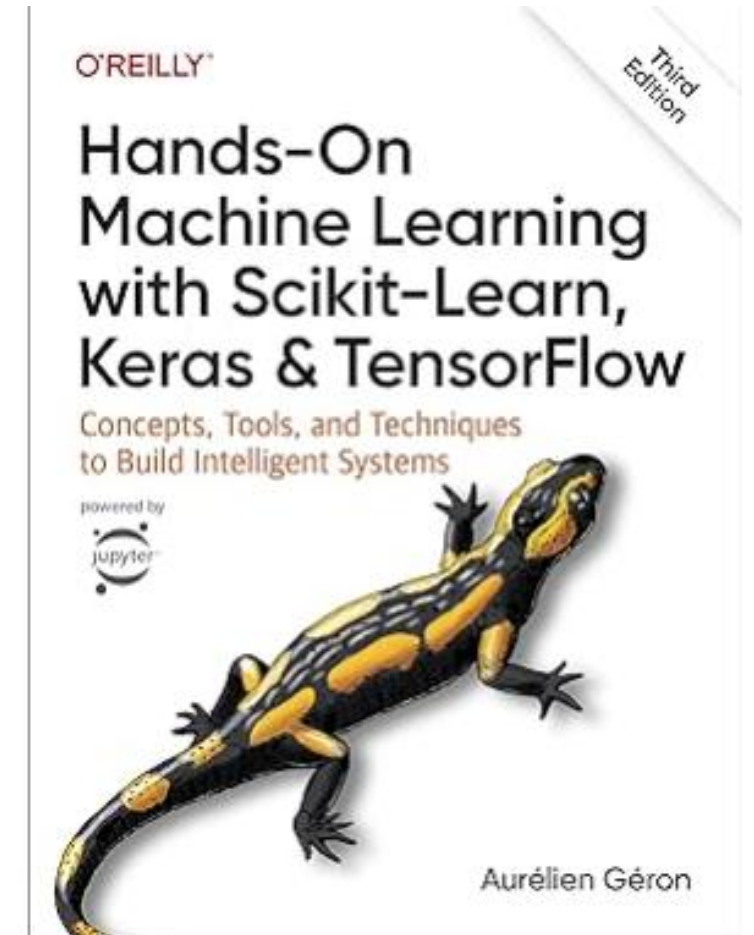
- My email address: avimohan@bu.edu
 - Office hours: by appointment.
 - Subject line: **[CS 767 Spring 2025]**
 - Including square brackets
 - Otherwise, no guarantee of response!
- TA: Rajesh Chandra (rj21@bu.edu)
 - Office hours: TBD

Course Schedule (***TENTATIVE!!***)

Lecture Number	Lecture date	Topics	Homework	Comments on lectures or homework	Readings/Misc.
1	Jan 21st	Discussion of Prerequisites, Homework Structure, Mid-term date etc. Introduction to statistical learning	HW 1	Linear, polynomial, and logistic regression	Chapter 1
2	Jan 28th	Review of Matrix Algebra, Tutorials: NumPy, and Matplotlib	HW 2	Questions on NumPy, Matplotlib and Matrix algebra	Link to Geron's tutorial on Matrix Algebra
3	Feb 4th	Review of statistics, and basic probability Tutorials: Seaborn, and Pandas	HW 3	Seaborn, Pandas, basic probability, statistics	<i>"One of the most crucial skills a data scientist needs to have is the ability to think probabilistically ... In [data science] interviews, probability's foundational concepts are heavily tested, particularly conditional probability and basic applications involving PDFs of various probability distributions." -- Kevin Huo and Nick Singh.</i>
4	Feb 11th	Decision Trees, Tutorial: Decision Trees in Scikit-Learn	HW 4	Simple dataset for Decision Trees	Chapter 6
	Feb 18th	Monday schedule (no class)			
5	Feb 25th	Boosting, Bagging, and Random Forests	HW 6	Bagging, Boosting, and Random Forests	Chapter 7: pp 220-232 (211 - 218 if time permits)
6	March 4th	Support Vector Machines Tutorial: SVMs in Sci-kit Learn	HW 7	SVMs	Chapter 5
7	March 11th	Optimization Algorithms (GD, SGD, RMSProp, and Adam)	HW 8	Optimization basics, Gradient descent <i>This topic could bleed into the next class.</i>	Link to Geron's tutorial on Differential Calculus
	March 18th	Spring Break (no class)			
	March 25th	Mid-term Exam			
8	April 1st	Feedforward Networks, Single-layer networks for regression and classification, Tutorial: Basics of TensorFlow	HW 8	Designing FF networks in TF, tracking the trajectory of SGD, choosing the right learning rate. Reading: 10 (Keras API), 11: pp 357 - 384 Chapter 12 is very useful.	Zachary Lipton's book has good TF tutorials E.g.: basic regression, classification.
	April 8th	Backpropagation	HW 9		
9	April 15th	Convolutional Neural Networks	HW 10	Simple computer vision related datasets	Chapter 14: pp 479 - 493, Time permitting: pp 499 (AlexNet), 505 (ResNets), 527 (YOLO)
10	April 22nd	Recurrent Neural Networks and LSTMs	HW 11	LSTMs, GRUs, and RNNs datasets related to time series analysis	Chapter 15 (will also be referring to Chapter 9 in Lipton's book)
11	April 29th	Advanced topics (self-attention, GANs, etc.)	<i>No homework, prepare for semester finals.</i>		Chapter 16, Chapter 17

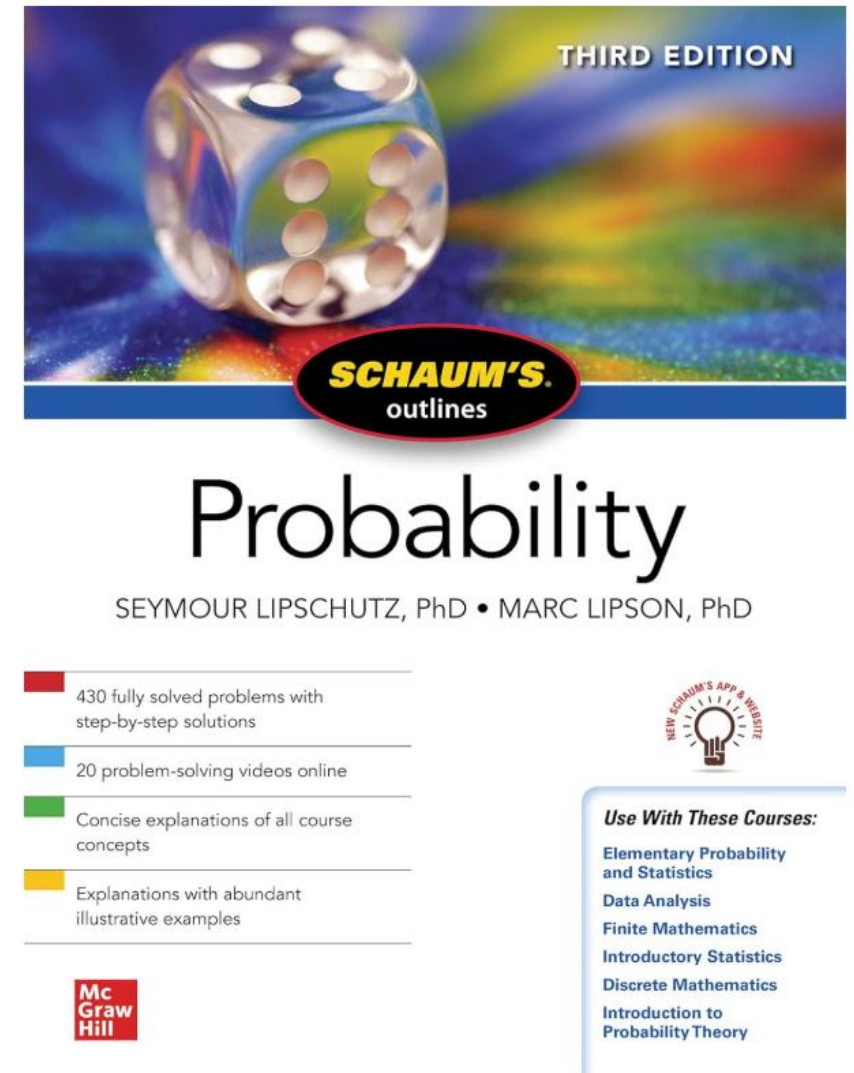
References

- Textbook: Géron, A., 2022. “***Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow.***” O'Reilly Media, Inc.
- I have reserved this book in the SEL library.
 - Please email either **Tracy Stone** (tstone@bu.edu), or **Sara Goodwin** (circulation@bu.libanswers.com) if you aren't able to access the books.
- <https://github.com/ageron/handson-ml3>



References

- Seymour Lipschutz, “***Probability,***” ***Schaum’s Outline of Probability***, 3rd Edition.
- Solved problems
- I have reserved this book in the SEL library.
 - Please email either **Tracy Stone** (tstone@bu.edu), or **Sara Goodwin** (circulation@bu.libanswers.com) if you aren’t able to access the books.



Other references

- Zhang, A., Lipton, Z.C., Li, M. and Smola, A.J., 2021. “***Dive Into Deep Learning.***”
 - Link: <https://d2l.ai/index.html>
- Ian Goodfellow, Yoshua Bengio, and Aaron Courville, “***Deep learning,***” The MIT Press, 2016.
 - Link: <https://www.deeplearningbook.org/>

Review of Fundamentals

Logistic Regression: algorithm flow

